



HABIB UNIVERSITY

DHANANI SCHOOL OF SCIENCE AND ENGINEERING

EE251/CE252	Spring'26	HW 3 SOLUTION	Marks 50 Weight 2.5%	Due: As announced
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Important Notes

Preceding the solutions, you must include the following information:

- Your name
- Student ID
- Course ID
- Homework number
- Date and time submitted

Solution should be elaborative, sketch neat circuits wherever required, show clearly all steps of calculation & mention reference (for example formula, law, rule) to strengthen your understanding to solve the electric circuit problems.

Also, please comply with Academic Integrity Policies of Habib University related to plagiarism while submitting this assignment for grading.

Late Submission Policy

As announced in course syllabus.

CLO-3	Analyze DT signals and systems in the transform domain using Z transform.	PLO-2 (Problem Analysis)	Cognitive Level-4 (Analyzing)
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Problem 1 [10 pts]

Solve the following difference equation using z-transforms:

$$y[n] + 3y[n-1] + 2y[n-2] = 2x[n] - x[n-1]; y[-1] = 0; y[-2] = 1, x[n] = u[n]$$

Problem 2 [10 pts]

Find the z transform of the following signals:

a) $x[n] = u[n] - u[n-4]$

b) $x[n] = 0.5^n u[n]$

c) $x[n] = \begin{matrix} \downarrow \\ [1 \ 4 \ 8 \ 2] \end{matrix}$

Problem 3 [10 pts]

Using only the fact that $\gamma^n u[n] \longleftrightarrow z/(z - \gamma)$ and properties of the z -transform, find the z -transform of each of the following:

- (a) $n^2 u[n]$
- (b) $n^2 \gamma^n u[n]$
- (c) $n^3 u[n]$
- (d) $a^n [u[n] - u[n - m]]$

Problem 4 [10 pts]

Consider a DT-LTI system given by the difference equation $y[n] - \frac{1}{4}y[n-2] = x[n-1]$

- (a) Use transform-domain techniques to determine the zero-state response $y_{\text{zsr}}[n]$ to input $x[n] = 3u[n-5]$.
- (b) Use transform-domain techniques to determine the zero-input response $y_{\text{zir}}[n]$ given $y_{\text{zir}}[-2] = y_{\text{zir}}[-1] = 1$.

Problem 5 [10 pts]

Consider transfer functions $H[z]$ of three different DT-LTI systems given below. For each system, find the poles and zeros and use these to determine the stability and invertibility of the systems.

- (a) $\frac{z(z-4)}{z^2-5z+6}$
 (b) $\frac{z-4}{z^2-5z+6}$
 (c) $\frac{(e^{-2}-2)z}{(z-e^{-2})(z-2)}$

Solution

Problem 1

$$y(z) + 3(z^{-1}y(z) + y[-1]) + 2(z^{-2}y(z) + y[-2] + z^{-1}y[-1]) \\ = 2x(z) - z^{-1}x(z) - x[-1]$$

$$y(z)(1+3z^{-1}+2z^{-2}) + 2 = \frac{2z}{z-1} - \frac{1}{z-1}$$

$$y(z) = \frac{z^2}{z^2+3z+2} \cdot \frac{2z-1}{z-1}$$

$$= \frac{z^2}{(z+1)(z+2)(z-1)}$$

$$= \frac{C_1}{z+1} + \frac{C_2}{z+2} + \frac{C_3}{z-1}$$

$$= \frac{-1/2}{z+1} + \frac{4/3}{z+2} + \frac{1/6}{z-1}$$

$$zy(z) = \frac{-1/2z}{z+1} + \frac{4/3z}{z+2} + \frac{1/6z}{z-1}$$

$$\longrightarrow \left(-\frac{1}{2}(1)^n + \frac{4}{3}(-2)^n + \frac{1}{6} \right) u[n]$$

$$y[n] = \left(-\frac{1}{2}(-1)^{n-1} + \frac{4}{3}(-2)^{n-1} + \frac{1}{6} \right) u[n-1]$$

Problem 2

$$a) \quad x[n] = u[n] - u[n-4]$$

$$X(z) = \sum_{n=0}^3 z^{-n} = 1 + z^{-1} + z^{-2} + z^{-3}$$

$$b) \quad x[n] = 0.5^n u[n]$$

$$\begin{aligned} X(z) &= \sum_{n=-\infty}^{\infty} 0.5^n u[n] z^{-n} \\ &= \sum_{n=0}^{\infty} (0.5 z^{-1})^n \end{aligned}$$

use geometric series convergence:

$$\sum_{n=0}^{\infty} a^n = \frac{1}{1-a} \quad \text{if } |a| < 1$$

$$X(z) = \frac{1}{1 - 0.5z^{-1}}$$

$$c) \quad x[n] = [1 \quad 4 \quad 8 \quad 2]$$

$$X(z) = 1 + 4z^{-1} + 8z^{-2} + 2z^{-3}$$

Problem 3:

(a) Consider

$$\gamma^n u[n] \iff \frac{z}{z-\gamma}.$$

Application of the multiplication property Eq. (5.18) to this pair yields

$$n\gamma^n u[n] \iff -z \frac{d}{dz} \left(\frac{z}{z-\gamma} \right) = \frac{\gamma z}{(z-\gamma)^2}.$$

One more application of the multiplication property yields

$$n^2 \gamma^n u[n] \iff -z \frac{d}{dz} \left(\frac{\gamma z}{(z-\gamma)^2} \right) = \frac{\gamma z(z+\gamma)}{(z-\gamma)^3}.$$

Letting $\gamma = 1$, we obtain

$$n^2 u[n] \iff \frac{z(z+1)}{(z-1)^3}.$$

(b) Consider

$$\gamma^n u[n] \iff \frac{z}{z-\gamma}.$$

Application of the multiplication property Eq. (5.18) to this pair yields

$$n\gamma^n u[n] \iff -z \frac{d}{dz} \left(\frac{z}{z-\gamma} \right) = \frac{\gamma z}{(z-\gamma)^2}.$$

One more application of the multiplication property yields

$$n^2 \gamma^n u[n] \iff -z \frac{d}{dz} \left(\frac{\gamma z}{(z-\gamma)^2} \right) = \frac{\gamma z(z+\gamma)}{(z-\gamma)^3}.$$

(c) Application of Eq. (5.18) to $n^2 \gamma^n u[n] \iff \frac{\gamma z(z+\gamma)}{(z-\gamma)^3}$ (found in part a) yields

$$n^3 \gamma^n u[n] = -z \frac{d}{dz} \left(\frac{\gamma z(z+\gamma)}{(z-\gamma)^3} \right) = \frac{\gamma z(z^2 + 4\gamma z + \gamma^2)}{(z-\gamma)^4}.$$

Setting $\gamma = 1$ in this result yields

$$n^3 u[n] = \frac{z(z^2 + 4z + 1)}{(z-1)^4}$$

Problem 4

(a) Taking the z -transform of the difference equation (0 ICs) yields

$$Y_{\text{zsr}}[z] - \frac{1}{4}z^{-2}Y_{\text{zsr}}[z] = z^{-1}X[z].$$

Since $x[n] = 3u[n-5]$, we see that

$$Y_{\text{zsr}}[z] \left(1 - \frac{1}{4}z^{-2}\right) = z^{-1}3z^{-5}\frac{z}{z-1}.$$

Solving for $Y_{\text{zsr}}[z]$ yields

$$Y_{\text{zsr}}[z] = \frac{3z^{-5}}{(z-1)(1-\frac{1}{2}z^{-1})(1+\frac{1}{2}z^{-1})} = \frac{3z^{-3}}{(z-1)(z-\frac{1}{2})(z+\frac{1}{2})}.$$

The modified partial fraction expansion is thus

$$\frac{Y_{\text{zsr}}[z]}{z} = 3z^{-4} \left(\frac{\frac{4}{3}}{z-1} + \frac{-2}{z-\frac{1}{2}} + \frac{\frac{2}{3}}{z+\frac{1}{2}} \right).$$

Inverting, we obtain

$$y_{\text{zsr}}[n] = \left(4 - 6\left(\frac{1}{2}\right)^{n-4} + 2\left(-\frac{1}{2}\right)^{n-4}\right) u[n-4].$$

(b) Taking the unilateral z -transform of the difference equation with $x[n] = 0$ yields

$$Y_{\text{zir}}[z] - \frac{1}{4} [z^{-2}Y_{\text{zir}}[z] + z^{-2}(y[-1]z^1 + y[-2]z^2)] = 0.$$

Substituting $y_{\text{zir}}[-2] = y_{\text{zir}}[-1] = 1$ and then rearranging yield

$$Y_{\text{zir}}[z] \left(1 - \frac{1}{4}z^{-2}\right) = \frac{1}{4} + \frac{1}{4}z^{-1}.$$

Thus,

$$\frac{Y_{\text{zir}}[z]}{z} = \frac{\frac{1}{4}(z+1)}{z^2 - \frac{1}{4}} = \frac{\frac{3}{8}}{z - \frac{1}{2}} + \frac{-\frac{1}{8}}{z + \frac{1}{2}}.$$

Inverting yields

$$y_{\text{zir}}[n] = \left(\frac{3}{8}\left(\frac{1}{2}\right)^n - \frac{1}{8}\left(-\frac{1}{2}\right)^n\right) u[n].$$

Problem 5

(a)

$$\frac{z(z-4)}{(z-2)(z-3)}$$

Zeroes: 0, 4

Poles: 2, 3

Poles outside unit circle – therefore system is unstable.

One of the zeroes is outside the unit circle – therefore system is not invertible.

(b)

$$\frac{(z - 4)}{(z - 2)(z - 3)}$$

Zeroes: 4

Poles: 2, 3

Poles outside unit circle – therefore system is unstable.

One of the zeroes is outside the unit circle – therefore system is not invertible.

(c) Obvious from given transfer function that

Zeroes: 0

Poles: e^{-2} , 2

Thus system is invertible but unstable.